SECI1013: DISCRETE STRUCTURES

SESSION 2025/2026 – SEMESTER 1

ASSIGNMENT 1 (CHAPTER 1 –SET THEORY & LOGIC)

INSTRUCTIONS:

a. This assignment must be conducted in a group. Please clearly write the group members' names & matric numbers on the front page of the submission.
b. Solutions for each question must be readable and neatly written on plain A4 paper. Every step or calculation should be properly shown. Failure to do so will result in the rejection of the submission of the assignment.
c. This assignment consists of 7 questions (60 marks), contributing 3% of overall course marks.

Question 1 **[6 marks]**

A university's student information system stores the student identification numbers of students registered for a programming workshop. Due to a system error, some student identification numbers appear more than once in the registration list.

The recorded identification numbers are:

`A102, A105, A103, A102, A108, A105, A110, A103`

Let (S) represent the set of distinct student identification numbers recorded in the system. Answer the following questions:

(a) Express (S) in complete list form. (1 mark)
(b) Determine the cardinality of (S). (1 mark)
(c) Determine whether (A105 $\in$ S) and (A107 $\in$ S). Justify your answers. (2 marks)
(d) Explain why the repeated identification numbers do not increase the cardinality of (S). (1 mark)
(e) Express (S) in set-builder notation using the condition that an element is a distinct student identification number in the recorded list. (1 mark)

Question 2 **[8 marks]**

A mobile security application performs a diagnostic scan on a user's phone to analyze privacy risks. The scanner categorizes mobile apps based on three key permissions they request:

- C: The set of apps requesting **Camera** access.
- L: The set of apps requesting **Location** services.
- M: The set of apps requesting **Microphone** access.

During a system audit of **50 installed applications**, the scan reports the following raw diagnostic data:

- **26** apps request Camera access ($|C| = 26$)
- **22** apps request Location services ($|L| = 22$)
- **18** apps request Microphone access ($|M| = 18$)
- **10** apps request both Camera and Location ($|C \cap L| = 10$)
- **8** apps request both Location and Microphone ($|L \cap M| = 8$)
- **7** apps request both Camera and Microphone ($|C \cap M| = 7$)
- **3** apps request all three permissions ($|C \cap L \cap M| = 3$)

Answer the following questions:

a) Calculate the total number of applications that request at least one of these three permissions. (2 marks)

b) Determine how many application on the device request no permission at all among these three categories that relative to the universal set U of all installed apps. (1 mark)

c) A security advisory flags apps that request camera access exclusively (no location or microphone) as potential background spyware. Write a set notation for this target group and calculate its cardinality. (2 marks)

d) Draw a venn diagram representing the breakdown of all 50 applications. (3 marks)

Question 3 **[6 marks]**

A university offers several computing courses and assigns students to different sections. Let the set of course be :

$$C = \{AI, DS, SE, CS\}$$

Where AI represents Artificial Intelligence, DS represents Data Science, SE. represents Software Engineering and CS represents Computer Science.

Let the set sections be :

S={1,2,3,4,5}

The university uses the cartesian product (C⊕ S) to present all possible course-section combinations.

Answer the following question

a) Express set (C) and (S) in set-buider notation (2 marks)

b) Determine the Cartesian product (C⊕ S) in set-buider notation (1 mark)

c) Determine the cardinality of (C⊕ S). (1 mark)

d) The university decides that only section groups 1,2 and 3 are available for Artificial Intelligence while all 5 sections are available for each of the other three courses. Define a set (R) containing all valid course-section group combinations. Express (R) in complete list (2 marks)

Question 4 **[12 marks]**

a) If $p \rightarrow q$ is false what is the truth value of $((\neg p) \wedge q) \leftrightarrow (p \vee q)$? (2 marks)
b) Construct the truth table for $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$ (6 marks)
c) Write these propositions using *p*, *q*, and *r* and logical connectives (including negations).
   i) You get an A on the final, you do every exercise in this book, and you get an A in this class. (2 ½ marks)
   ii) To get an A in this class, it is necessary for you to get an A on the final. (1 ½ marks)

Question 5 **[8 marks]**

Express the following sentence into propositional function, quantifiers and logical connective.
a) A passenger on an airline qualifies as an elite flyer if the passenger flies more than 25,000 miles in a year or takes more than 25 flights during that year. Domain of discourse is all people. (3 marks)
b) There is a student who has taken more than 21 credit hours in a semester and received all A's. Domain of discourse is all students. (2 marks)
c) Negate the logical expression in (b) and write the sentence. (2 marks)

Question 6 **[8 marks]**

a) Let $Q(x, y)$ be the statement "$x + y = x - y$." If the domain for both variables consists of all integers, what are the truth values and explain?
   i. $\forall x \exists y\, Q(x, y)$ (2 marks)
   ii. Express the negation of (i) and state the truth value of the negation statement. (2 marks)

b) Let $M(x, y)$ be "x has sent y an e-mail message" and $T(x, y)$ be "x has telephoned y," where the domain consists of all students in your class. Use quantifiers to express each of these statements.
   i. There is a student in your class who has sent everyone else in your class an e-mail message. (2 marks)
   ii. There is someone in your class who has either sent an e-mail message or telephoned everyone else in your class. (2 marks)

Question 7 **[12 marks]**

a) Use contrapositive proof to show, if $(x^2 + 1)(y + 1)$ is an odd then x is odd integer or *y* is an odd integer. (4 marks)
b) Use contradict proof to show for all integers a and b if $a^2 + b^2$ is odd, then a or b is odd. (4 marks)
c) Use direct proof to show for all integers n if n is a multiple of 3, then n can be written as the sum consecutive integer. (4 marks)